

EDUCATIONAL PERILUNE

Visual Effects Educational Practices at Media Design School

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ABSTRACT

This work introduces and explores learning experience design (LXD) methods to teaching and learning in the creative technologies space. We examine LXD techniques as a method to provide educational solutions to the application of knowledge and skills in project-based learning with the goal of creating an engaging, interactive, rich, and effective learning environment. We analyse LXD techniques implemented in instruction designed specifically for CGI, 3D animation and visual effects subject matter.

KEYWORDS

project based learning, experiential learning, learning experience design

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1 INTRODUCTION

Media Design School (MDS) is a provider of education in Auckland, rated by the New Zealand Qualifications Authority as Category One; Excellence in educational performance and self-assessment. From its inception in 1998, MDS programmes have practiced focused, sequential learning with scaffolded components, each subject building on the knowledge developed in a cumulative manner from previous subjects. Project based learning is another founding principle of the institution assessing learning through the learner's ability to apply knowledge to creative outputs. MDS aims to provide exceptional graduates who immediately add value to the well-established digital and creative technology industry, a clear mission that drives our strategy and educational design. The work presented here will focus on the Bachelor of Art and Design (3D Animation and Visual Effects) Bachelor's degree programme and advancements in teaching this subject matter.

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The model offered by many animation and visual effects education providers is delivered through a mixture of theoretical and practical lectures followed by project-based assessment assignments. We apply these methods in the first two years of the MDS, Bachelor of Art and Design (BoAD) programme. These models of delivery and assessment are derived from an intrinsically cognitive and behavioural teaching and learning approach, intended to provide the basic principles and tacit knowledge required to achieve tasks while also concurrently developing critical thinking and problem-solving skills [Hilbert, 2015].

However, these methods separate the roles of teacher and learner and limit the engagement between the two parties. This creates an experience that can be cold, isolating and can reduce the process of learning to a purely academic exercise. Our objective is to break this barrier and create an engaging environment that immerses our students in a collaborative, meaningful experience which also prepares them for success after graduation. The 2022 film, *Perilune* is an example of the of a wholistic educational activity at MDS that transcends the normal classroom experience in a student's third and final year of study (Fig.1).

1.1 DESIGN AND IMPLIMENTATION

Review and self-assessment are integral to continual improvement for an educational provider. Project based-learning is proven to provide a rich learning environment but is dependent on engagement. The process of building upon the successes of project-based learning with the objective of increasing student engagement led to the examination of emerging educational techniques which support student and stakeholder needs. Learning Experience Design (LXD) engages learners through experiences. The needs and preferences of learners, as well as subject matter, dictate the design of the experience which ultimately will promote the acquisition and retention of knowledge and skills. As developed and defined by Niels Floor, LXD is “the process of creating learning experiences that enable the learner to achieve the desired learning outcome in a human centered and goal-oriented way” [Floor, 2023]. Reflecting on the needs of stakeholders including students, their families, industry, society, government, and the organization's aim to produce exceptional graduates inspired teaching methods designed around the LXD framework.

The emphasis on a goal-oriented method of engaging learners gave a starting point in the design of the experiential learning environment. Aligning the goals of stakeholders with a particular focus on visual effects (VFX) industry processes, and working back from

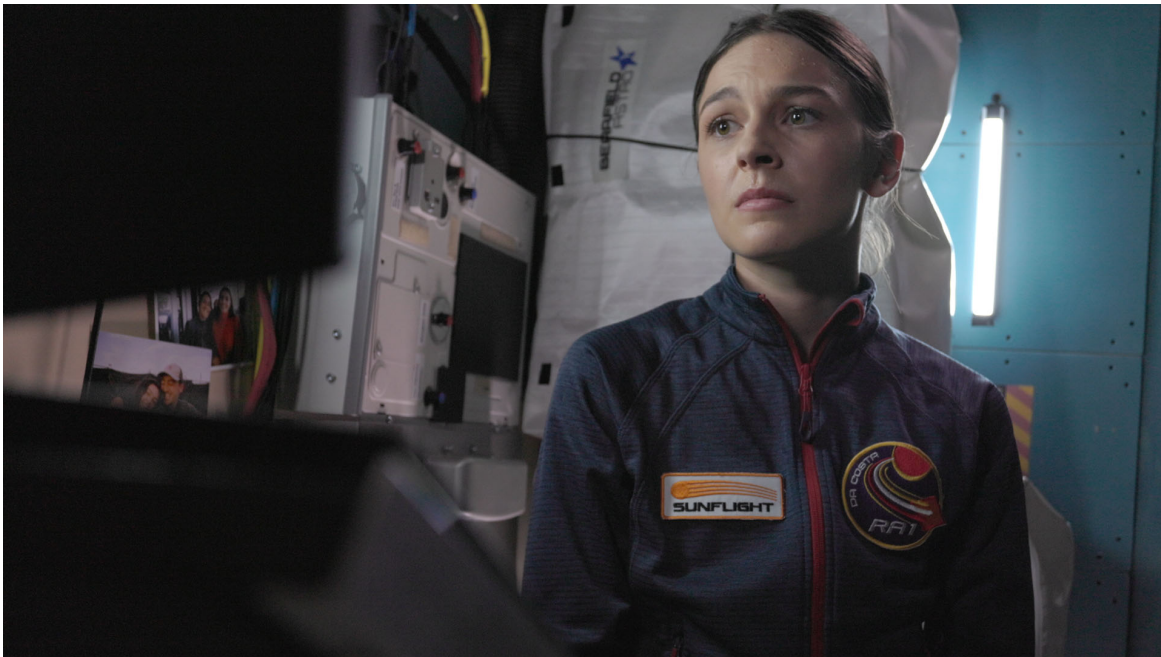


Figure 1: Completed VFX shot from Perilune

that point, produced an obvious solution. The proposition seems simple; to involve students in VFX productions in a professional environment with the goal to accomplish the release of a short film. The working VFX studio became the model on which learning design decisions are based. Students work collaboratively with faculty and industry professionals. Production pipeline methods are employed and learners work in interdisciplinary teams mimicking established industry workflows. However, it is important to note that this is not an internship programme and emphasis must remain focused on the process of learning as much as maintaining the real-world workplace experience.

In, *Experiential Learning: Experience as the Source of Learning and Development* [Kolb, 2014], Kolb defined learning as “the process whereby knowledge is created through the transformation of experience” (p. 38). Kolb identifies four essential stages of the learning experience consisting of: Concrete Experience, Reflective Observation, Abstract Conceptualization and Active Experimentation. Adhering to this framework ensures that the design of the learning experience follows accepted educational philosophies. Throughout the production exercise the learner engages in each stage transforming the experience into knowledge.

1-Concrete Experience. Students are involved in the early in pre-production through principal photography. Depending on learner preference and need, students participate in various aspects of the project on an extracurricular basis (Fig.2).

2-Reflective Observation. Following initial involvement learners feel a sense of ownership. The active engagement with the project allows students to reflect on what they have learned previously in their studies and prepare for the tasks to come.

3-Abstract conceptualization. As a member of the VFX team, learners now begin to understand the concepts underpinning the

pre-production and principle photographic phases of the project. Connections between abstract theories of filmmaking procedures taught in lectures and the experience they have participated in form as a basis of planning for the next phase of the experience.

4-Active experimentation. Learners now begin to actively apply learnings delivered in the first two years of their studies to the project that they have helped shape. Learners see the project as a whole and understand how their work fits into the overall production. Students must make connections between what they have experienced and what they already know to complete tasks in shot assignments, enhancing critical thinking and problem-solving skills (Fig.3).

ShotGrid project management software assists faculty in monitoring learners’ achievements and difficulties. In a dailies format, students must report on; what they achieved yesterday, what they plan to achieve today and what problems the learner is facing.

1.2 OUTCOMES

The BoAD programme has employed project-based learning in the form of VFX short films production of for over a decade. The learning benefit of working in a studio environment is undeniable. The knowledge and skills far outreach development of CGI and VFX technical and creative skills. Industry stakeholders emphasize the importance of soft skills in their workforce. As students engage in productions experiences they develop and improve their communication, collaboration and conflict resolution skills, their ability to take and utilize direction and critique to improve their work, recognize and meet production standards and develop confidence. The student engaged in the experience is no longer working in isolation on a project but is a member of a team with clearly defined



Figure 2: Students and crew shooting Perilune

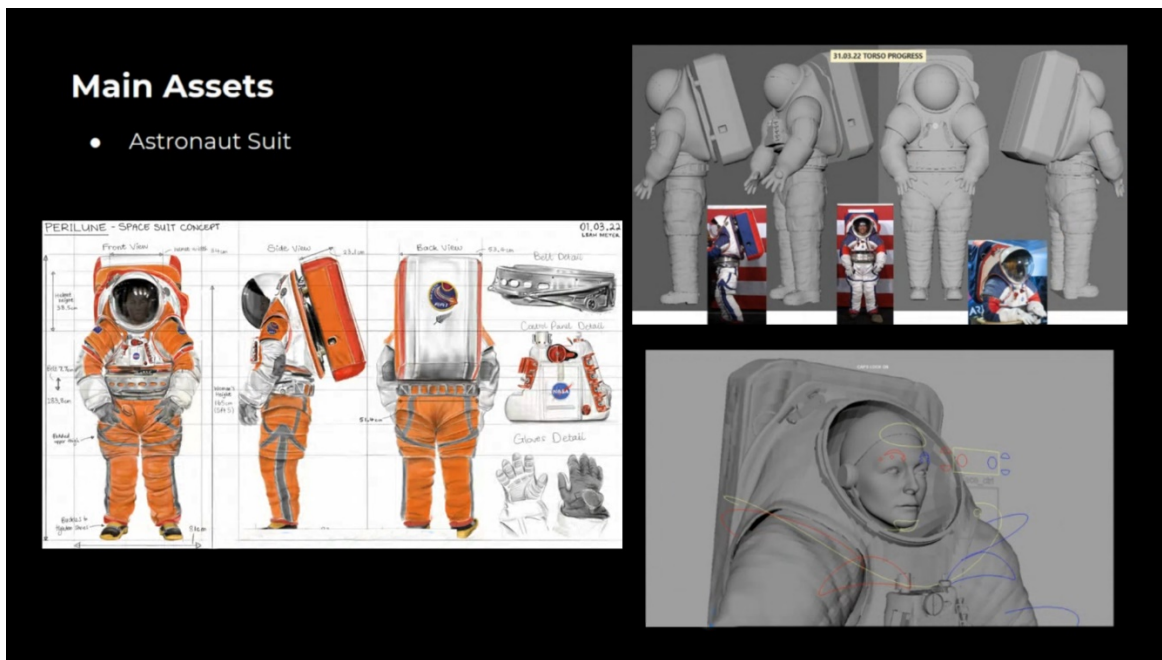


Figure 3: Development of CGI assets for Perilune

goals and common objectives. The environment in which they learn becomes far richer than the narrow tasks at hand.

The effectiveness of each project-based learning exercise is contingent on the design of the experience. The design of each experience must take into consideration the unique set of needs and preferences of the cohort of learners. This requires an agile approach to teaching and learning. Ridged curriculums with standardized lesson

plans, practiced and perfected over time do not work in this model. However, the LXD delivery model is not cost effective due to time requirements for design and preparation for bespoke educational solutions. The LXD oriented delivery model has tangible outcomes for students who are engaged and participate unreservedly that go far beyond meeting specified learning outcomes but can be challenging for educators and educational organizations to sustainably support.

1.3 CONCLUSION

Successful engagement is evident in an analysis of completion rates for subjects employing LXD. 97% of the cohort successfully completed Team Production 1, the Perilune VFX production component. Completion rates vary from 93% to 100% for cohorts studying in 2019 through 2021. Quality is another indicator of successful engagement. Perilune has been selected for screening in six international film festivals and has won the award for Best Director

– Short Animation at the 2023 Golden State Film Festival. Additionally, production experience on an award winning short film gives graduates an edge over graduates from other institutions in gaining employment and retaining those roles. A survey conducted in 2022 found that 72% of BoAD graduates had found employment in a related field within six months. 95% of BoAD graduates who completed their studies three years earlier were employed in a related field. Graduates are primed for a successful career leaving the programme with production experience on their CV, credits in an award-winning short film such as Perilune and the experienced gained at MDS.

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